APPLICATION OF PROCESS MINING IN PROJECT MANAGEMENT CONTEXT: AN EXPLORATORY CASE STUDY IN SOFTWARE DEVELOPMENT PROJECTS

Marco Antonio Archanjo
Rafael Araujo Kluska
Fernando Deschamps
Eduardo Alves Portela Santos

Introduction

Project: temporary endeavor undertaken to create a unique product, service or result

- Projects happen nowadays in dynamic, complex and fast changing environments
- A significant amount of data is collected, analyzed and transformed, becoming a valuable asset for organizations
- Due to the increased amount of data and information that needs to be integrated and communicated, a trend in project management is the use of automated tools able to collect, analyze and transform information

Introduction

According to Kerzner (2015), much of the success of current and future project managers depend on the infrastructure available (hardware and software):

This infrastructure will help:

- store larger amounts of generated project data
- control the multisourced flow of information
- process it
- make decisions based on it
- communicate it out to stakeholders

Introduction

Whyte J. et al. (2015):

- In the Big Data era, information becomes more and more an important organization asset and is considered a project delivery
- Information can be mined, combined and reused in different new manners.

Larson & Chang (2016):

- Project documentation is valuable, but has an intrinsic problem: usability
- Documentation requires a lot of time to be completed but tends to be out of date
- Usage of tools able to collect, process and show information quickly can aid in the decision making process

Swanson (2017):

- Artificial intelligence will help reduce human mistakes in project cost and scheduling estimates
- Machine Learning will be used to analyze project historical data and aid project manager make better decisions

Introduction

Process Mining: emerging method to explore historical data stored in information systems. It sits between data mining and computational intelligence on one side and process modelling and analysis on the other side

Process Mining Objectives: discover, monitor and improve real processes by extracting knowledge from event logs available in today's information systems:

- extracting process models from an event log
- conformance checking (monitoring deviations by comparing model and log)
- process enhancement (case prediction and history-based recommendations)
- social network/organizational mining

Introduction

Process Mining: already applied as a new tools in different areas:

- Healthcare (Mans, Schonenberg, Song, van der Aalst, & Bakker, 2008)
- Auditing (van der Aalst, van Hee, van Werf, & Verdonk, 2010)
- Fraud mitigation (Jans, Van Der Werf, Lybaert, & Vanhoof, 2011)
- Software process assessment (Valle, M. A., 2015)

Es-Soufi et al. (2016):

Process Mining combined with Machine Learning has been receiving much attention from Project Management and Product Development communities

Research Question

Based on the fact that:

- The usage of historical data is more relevant each day to improve project management practices in fast changing environments
- Increasing need to collect, compile, process, analyze and make information available in a comprehensive, quick and automated manner

The research question is formulated:

How can Process Mining be used to extract knowledge from the analysis of historical data present in project event logs created throughout the project life cycle?

Case study methodology with exploratory approach was chosen to develop this work. It contains techniques and resources that assure the quality and exemption in the proceedings.

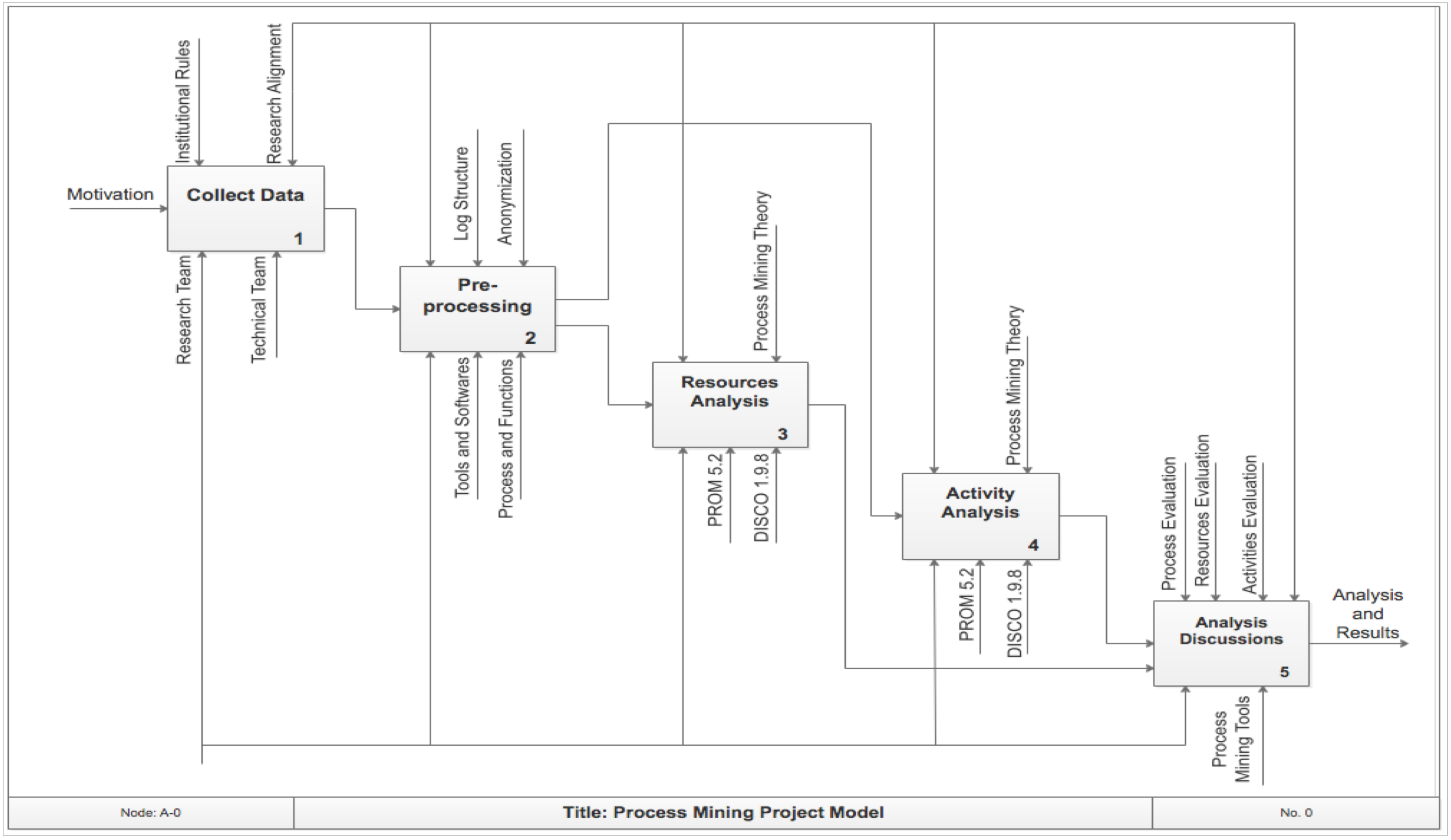
Process Mining Project Model

- Process Mining Project Model proposed by Kluska, Archanjo, Deschamps, and Santos (2017);

- Based on different techniques from different sources: Medeiros and Weijters (2008), van der Heijden (2012), van der Aalst W.M.P. et al. (2012) and van Eck, Lu, Leemans, and van der Aalst (2015);

- Documented in IDEF0 to create a clear description of the steps to be followed.

Process Mining Project Model



Data Collection

The Process Mining Project Model previously depicted was applied to logged events of seven software development projects of a CMMI Level 3 software development company located in Curitiba, Brazil.

Data Collection: registered event log files of the seven projects in XLSX format were supplied by the software development company. They were divided as described below:

- One XLSX file containing six projects;
- One XLSX file containing one project.

Pre-processing

Projects were grouped and the diversity of information in each logged file was evident. Two important activities needed to be performed before application of the Process Mining techniques:

- **Standardization** of the information before proceeding with process mining activities is a key step;
- **Anonymization** of the data registered in the event log files to preserve confidentiality.

The standardized and anonymous structure of the pre-processed files looked as the table below:

Index	Description
Case ID	Project identifier
TimeStamp - Start	Activity start date
TimeStamp - End	Activity finish date
Activity	Name of the activity
Resource	Resource assigned to the activity
Other	Other attributes that can be analyzed independent of the activity
Remove	Attribute to be removed from the analysis

Resource and Activity Analysis

- **Resource Analysis:**

  The following parameters were extracted using PROM5.2:
  - Log summary: contains the total number of events, total number of activities (event classes) and number of resources (originators) involved
  - Start Date
  - End Date
  - Distribution of activities among the different originators (resources) involved in the project
  - The Social Network Analysis
- **Activity Analysis:**
  - PROM5.2 plug-in named Heuristic Miner was used to mine the development process that was represented by a causal model
  - The causal model details the information flow and execution sequence of activities, however not taking resources into account

In this work, Project 7 was chosen to exemplify the application of the proposed Process Mining Project Model due to its reduced number of activities.

Project 7 - Resource Analysis

Project 7, despite the number of events (1592) and number of resources involved (19), has a reduced number of activities (14), which makes it a good example.

Project 7	Log Summary
Events	1592
Event Cases	14
Originators	19
Start Date	03/01/2016
End Date	09/12/2016

Originators	Occurrence (absolute)	Occurrence (relative)
Resource 2	360	22.613%
Resource13	222	13.945%
Resource 19	160	10.05%
Resource 10	140	8.794%
Resource 12	118	7.412%

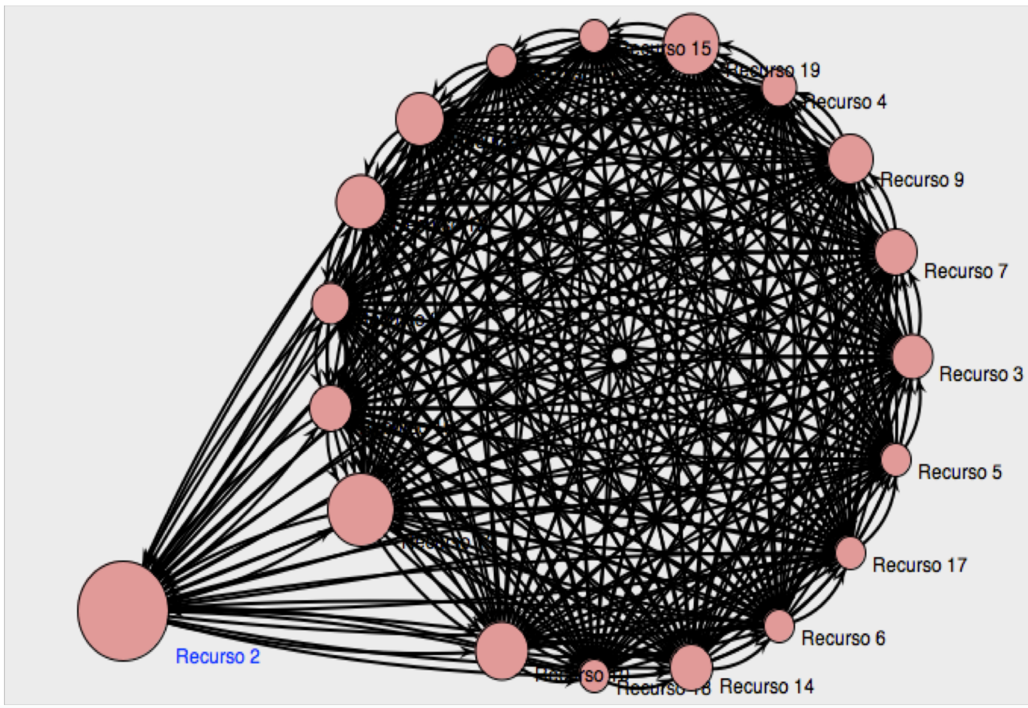
Analysis – Originator by Task Matrix

originator	Análise	Desenho	Desenvolvimento	Planejamento	Teste Homologa...	Teste Integrado	Teste Unitário
Recurso 1	6	0	46	5	0	0	0
Recurso 10	19	0	51	0	0	0	0
Recurso 11	0	0	0	2	0	0	0
Recurso 12	7	0	49	0	0	3	0
Recurso 13	25	0	84	2	0	0	0
Recurso 14	20	0	17	0	0	0	0
Recurso 15	2	0	0	0	0	0	0
Recurso 16	1	0	33	2	0	0	0
Recurso 17	0	0	0	1	0	1	0
Recurso 18	0	0	1	0	0	0	0
Recurso 19	26	1	49	0	4	0	0
Recurso 2	0	0	1	0	139	40	0
Recurso 3	1	1	30	0	0	0	0
Recurso 4	0	0	15	0	0	0	0
Recurso 5	0	0	0	0	0	2	0
Recurso 6	0	0	0	0	0	1	0
Recurso 7	9	0	29	0	0	0	0
Recurso 8	0	0	16	0	0	0	7
Recurso 9	48	0	0	0	0	0	0

Project 7 - Resource Analysis

Social Network Analysis:

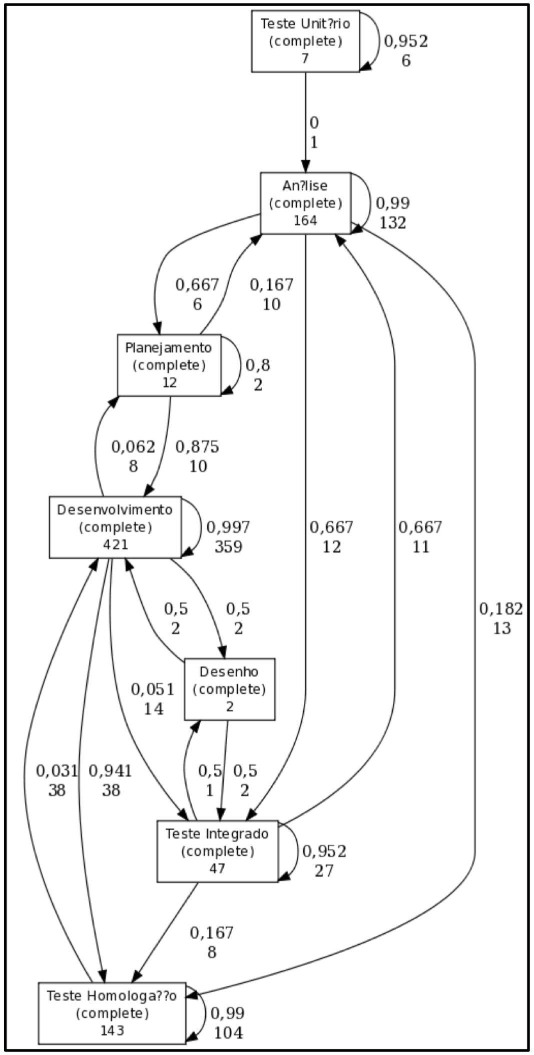
- Emphasizes the importance of resource 2 to Project 7, highlighted by the bigger circle representing resource 2 in the picture below
- Corroborates with the Originator by Task Matrix
- High number of handovers between the resources. That could be clarified by interviewing team members to better understand how the information flow worked during the project



Project 7 – Activity Analysis

Causal Model:

- The fact of having a big amount of events (1592) but a reduced number of activities (14) reflects directly in the causal model mined using PROM5.2 Heuristic Miner plug-in
- The causal model represents the sequence of activities performed in Project 7

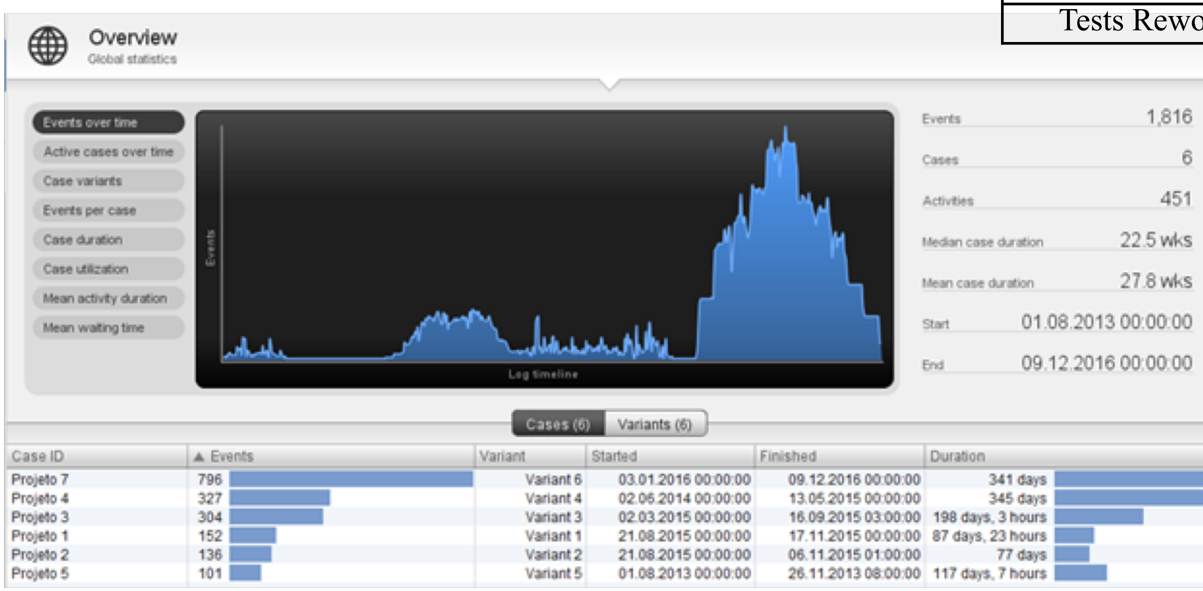


Combined Data – Activity Analysis

- Another interesting activity analysis that can be done using DISCO 1.9.8 is the distribution of events over time of the combined project file
- The combined project is a dataset composed by all data of six projects supplied by software development company

Applying DISCO 1.9.8 also reveals the mean duration and the duration range of the main activities performed in all six projects present in the combined project file.

Event Case	Mean Duration	Duration Range
Development	51 days, 3 hours	236 days, 17 hours
Analysis	53 days, 12 hours	265 days, 23 hours
Homologation Tests	24 days, 18 hours	207 days, 1 hour
Integration Tests	57 days, 23 hours	235 days, 23 hours
Tests Rework	8 days, 4 hours	98 days, 1 hour



During 2015, four projects were running in parallel, taking the number of events under execution to its maximum.

Conclusions

This case study had an exploratory approach brought forward different perspectives on resources, activities and time of software development projects

Resources perspective:

- Interaction between the different actors were synthesized through social network diagrams
- Handovers were mapped and the main resources could be clearly identified
- This kind of information could be used as lessons learned for future projects with similar characteristics and also useful for an ongoing project
- Bottlenecks related to resources could be identified and countermeasures could be taken to improve efficiency

Activities perspective:

- Causal models revealed the real sequence of activities performed in the project
- Causal models can be a starting point in planning new projects considering historical data from the very beginning
- The sequence of activities performed by the main resources involved in the project could also be mined. The link between main resource and activities performed by this person could be used to identify new team members with the same competences

Time perspective:

- Possible to extract the mean duration and the range duration of the activities when combining projects
- Information could be used for time estimation and in risk assessment of future projects

Thank you!